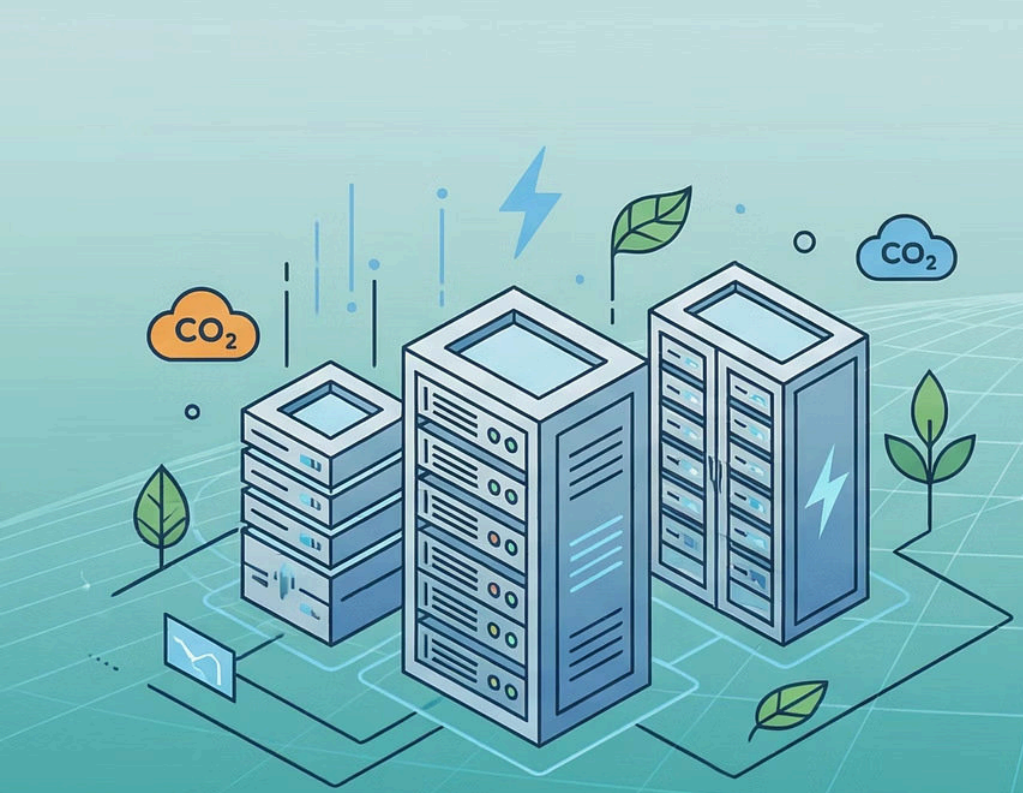
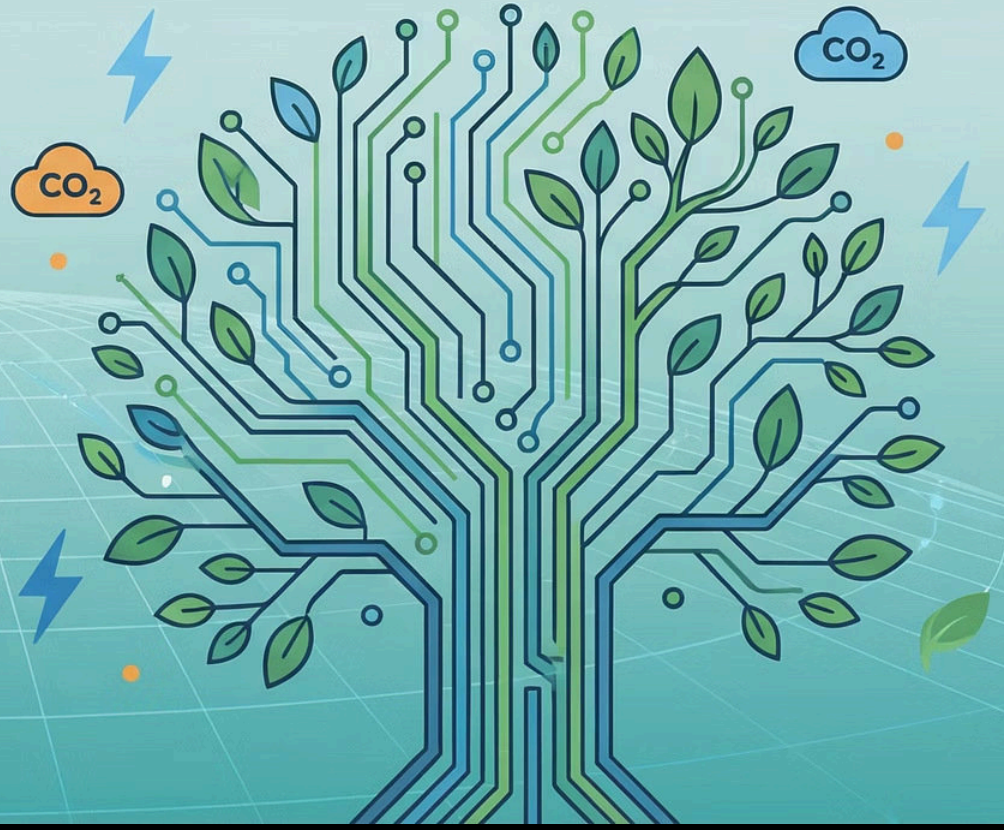
自然语言处理中的大模型训练与碳排放

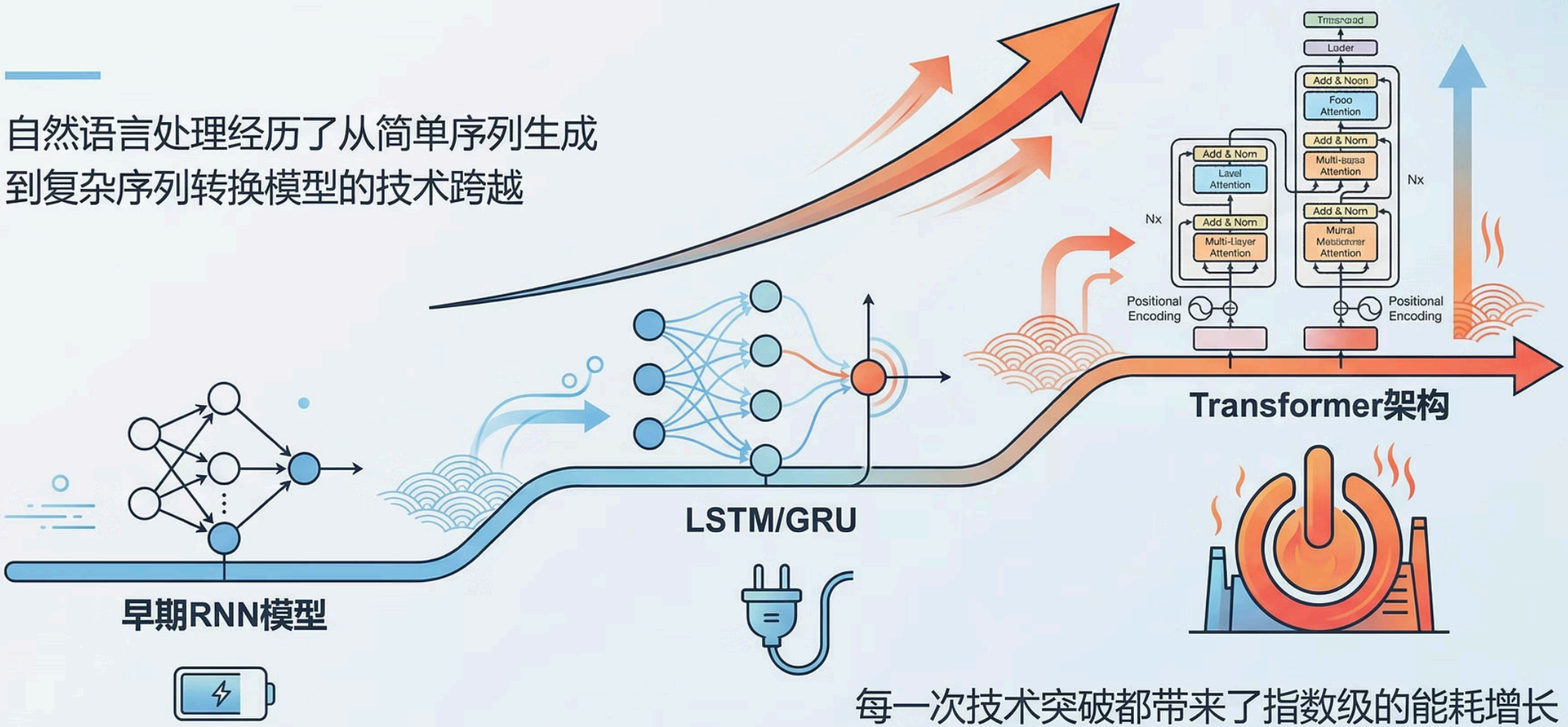
AI发展与环境责任的平衡之道





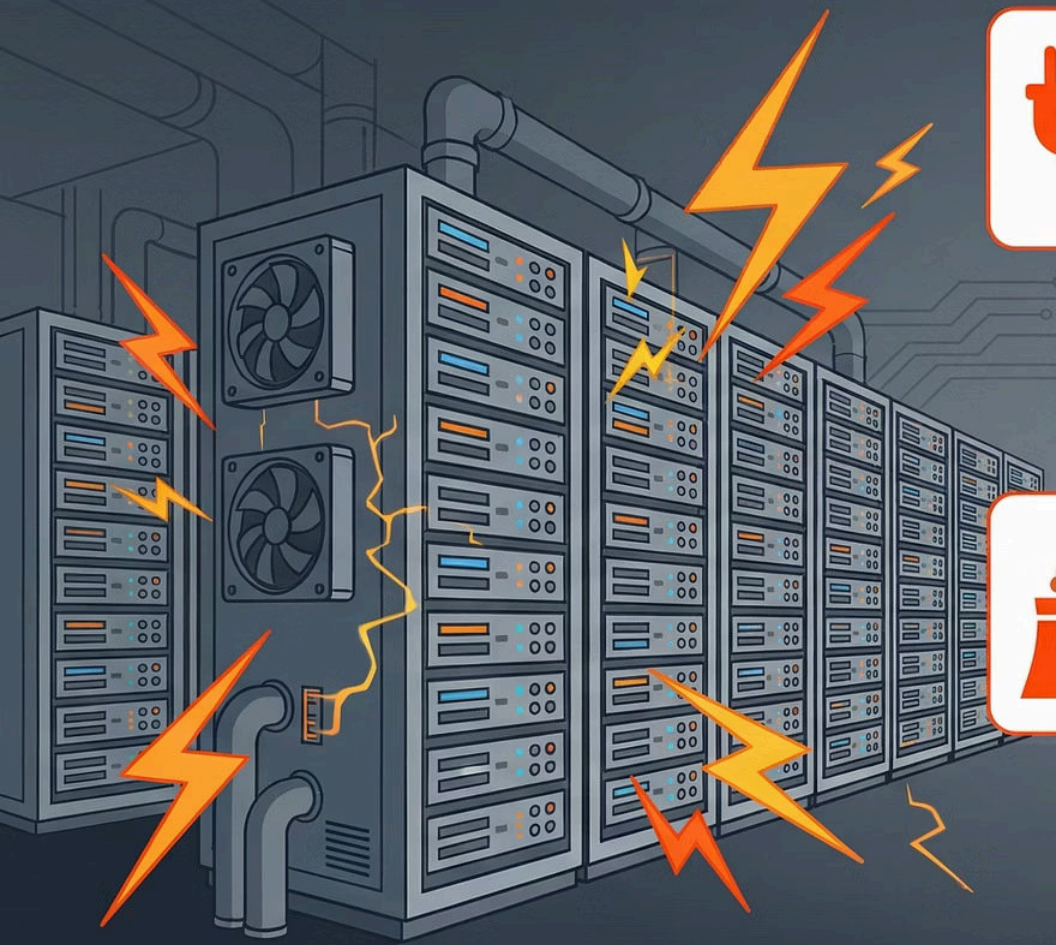
AI发展的能耗危机

自然语言处理经历了从简单序列生成到复杂序列转换模型的技术跨越



每一次技术突破都带来了指数级的能耗增长

GPT-3训练的惊人能耗



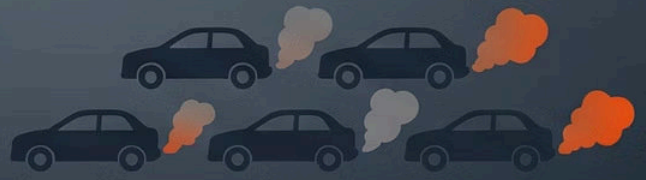
1287 MWh

120个美国家庭年用电量



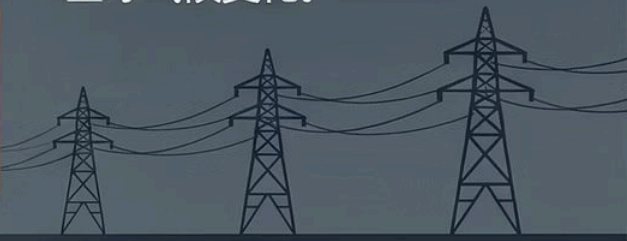
552吨CO_2

等同于5辆汽车终身排放



高能耗运算： 训练GPT-3这样的大型语言模型需要巨大的算力，其用电量相当于一座小型发电厂的短期产出。

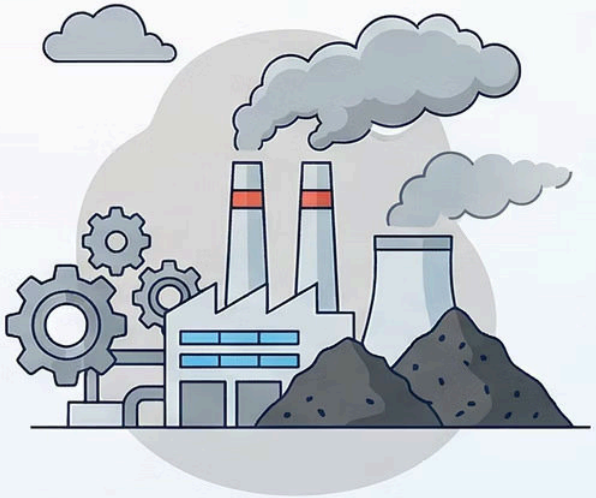
碳足迹影响： 巨额电力消耗转化为显著的碳排放，加剧全球气候变化。

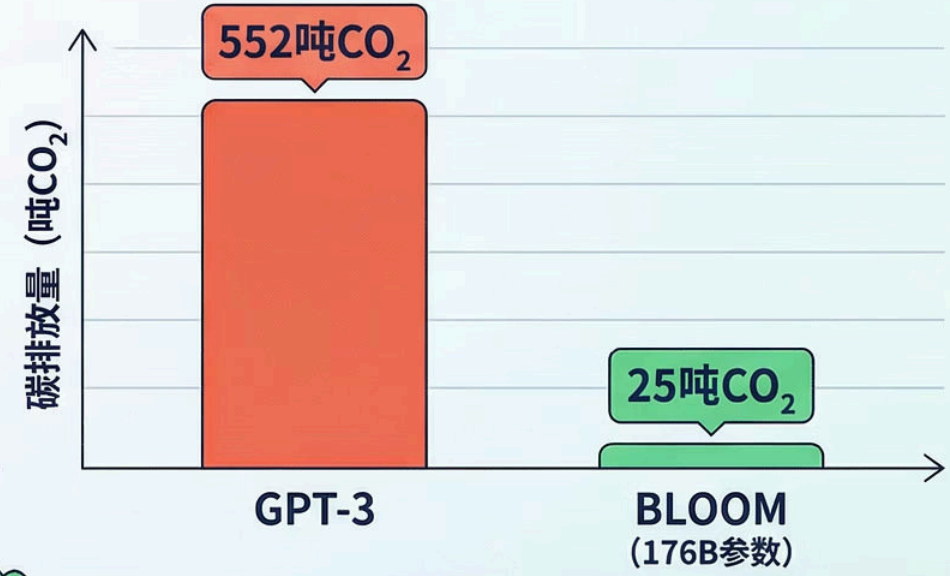


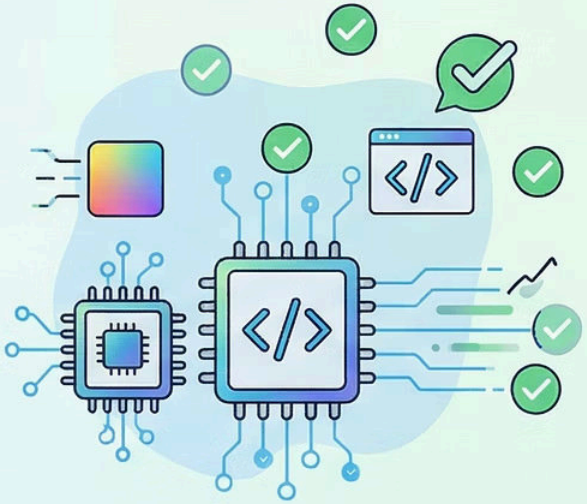
环境挑战： 人工智能的发展必须考虑其环境代价，寻找更可持续的训练方法和能源来源至关重要。

BLOOM模型的相对优势

BLOOM（176B参数）训练仅产生25吨CO_2排放



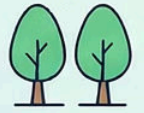






许多树木来抵消

证明了优化算法和架构设计的重要性



仅需少量树木

优化的架构设计和高效的硬件使用大幅降低了能耗

优化的架构设计和高效的硬件使用遛古成扶MB雕据持续大幅降低了能耗的功能，
其他量应用系统，就能出破行优化设计的架构转构流褶

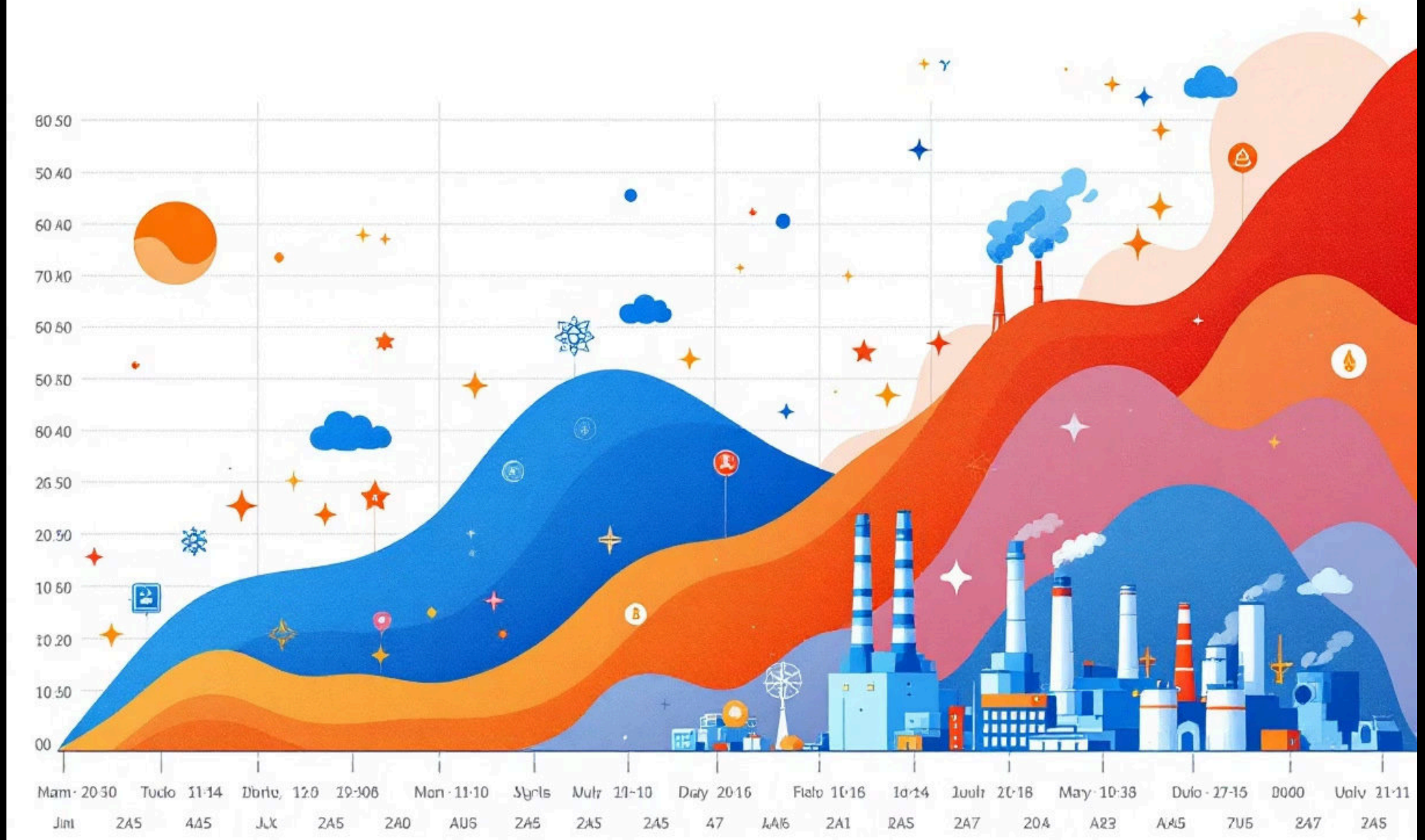
矛盾的核心



人工智能快速发展的迫切需求与
全球碳中和目标形成尖锐矛盾

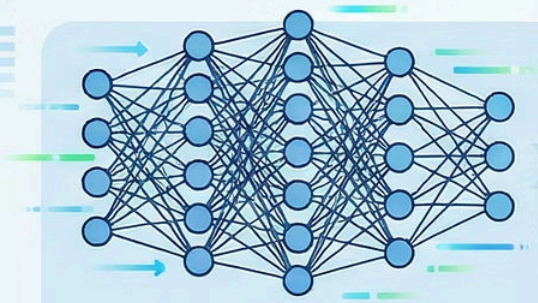
大模型性能提升往往以巨大
环境代价为前提

balancing technological progress with environmental responsibility.



‘绿色AI技术突破’

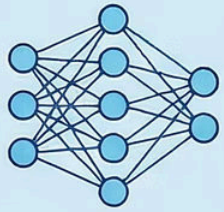
模型压缩技术的三大法宝



Teacher Model

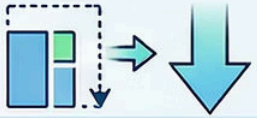


知识蒸馏：大模型→小模型

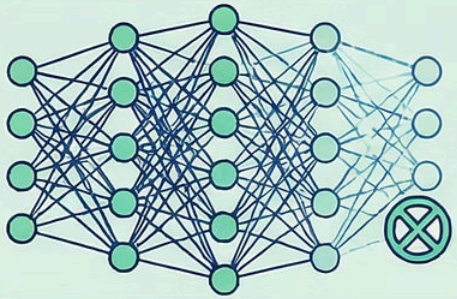


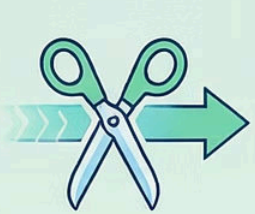
Student Model

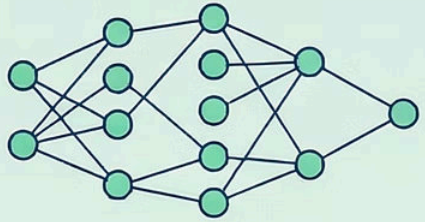
Teacher model shares its learned representations with the student, allowing the smaller model to mimic its performance without the computational burden. Knowledge transfer mechanics involve transferring soft targets and intermediate feature maps, significantly reducing complexity.



减少参数量：模型大小显著减小，存储更高效。



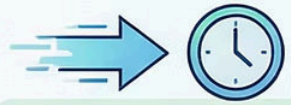




Pruned Model

网络剪枝：移除冗余参数

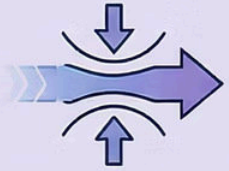
Redundancy removal benefits include identifying and eliminating unnecessary neurons and synapses, leading to significantly smaller model size, faster inference speeds, and lower power consumption without compromising accuracy.



更快推理：计算延迟降低，实时响应更迅速。



High Precision

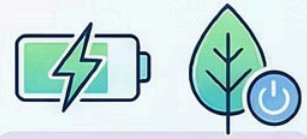




Low Precision

量化：FP32→INT8降低计算量

Quantization reduces the number of bits required to represent weights and activations, lowering memory footprint and enabling efficient arithmetic operations on specialized hardware, thus reducing computational complexity and energy usage.

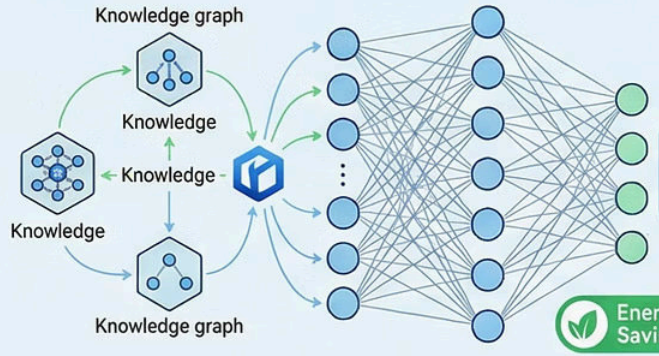


降低能耗：能效比大幅提升，更加环保节能。

生成此图像时出现错误

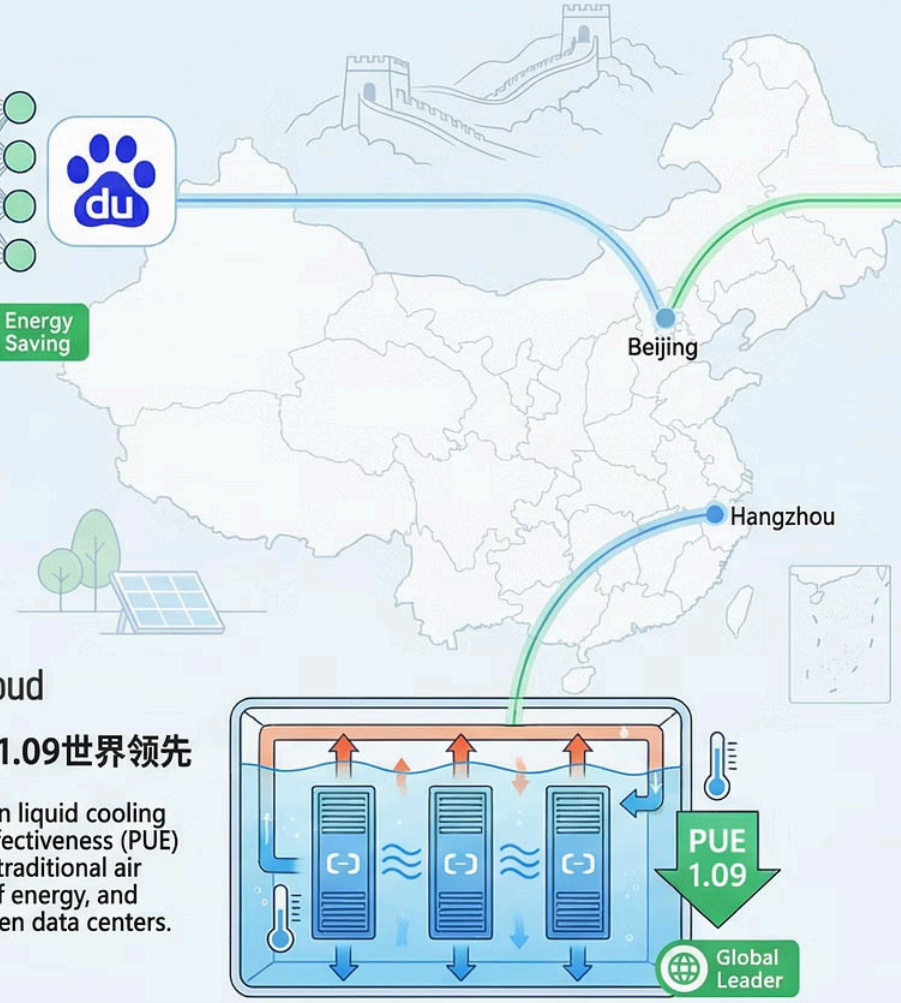
中国AI的绿色实践

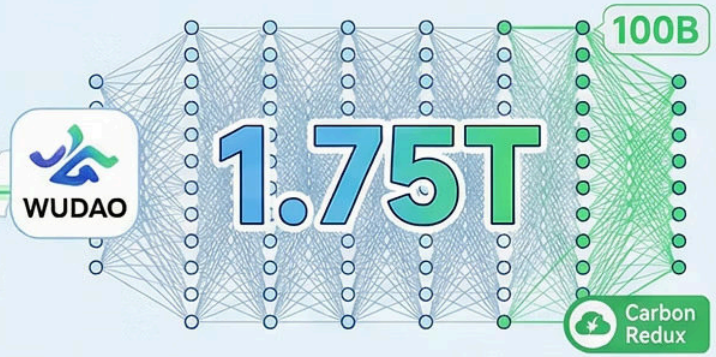
Leading sustainable innovation with advanced AI and eco-friendly infrastructure.



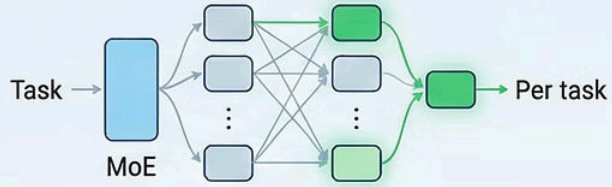
百度文心：知识增强技术降低参数需求

Through deep knowledge integration and enhanced pre-training, Baidu's Wenxin model significantly reduces parameter demands and computational resources, offering high performance with greater energy efficiency and lower carbon footprint.





智源悟道：1.75万亿参数，实际激活仅千亿级



智源悟道：1.75万亿参数，实际激活仅千亿级

Using Mixture of Experts (MoE) architecture, WuDao achieves massive scale (1.75 trillion parameters) while selectively activating only about 100 billion per task, drastically cutting down energy consumption and improving computational speed.

Alibaba Cloud

阿里云液冷技术：PUE降至1.09世界领先

Alibaba Cloud's innovative immersion liquid cooling technology reduces Power Usage Effectiveness (PUE) to a world-leading 1.09, eliminating traditional air conditioning, saving vast amounts of energy, and setting new global standards for green data centers.

环保责任与国家战略

AI发展绝不能以环境破坏为代价，这是时代的要求

We must ensure AI advancement does not come at the expense of our planet. This is the requirement of our era.

积极响应国家双碳战略：2030年碳达峰，2060年碳中和

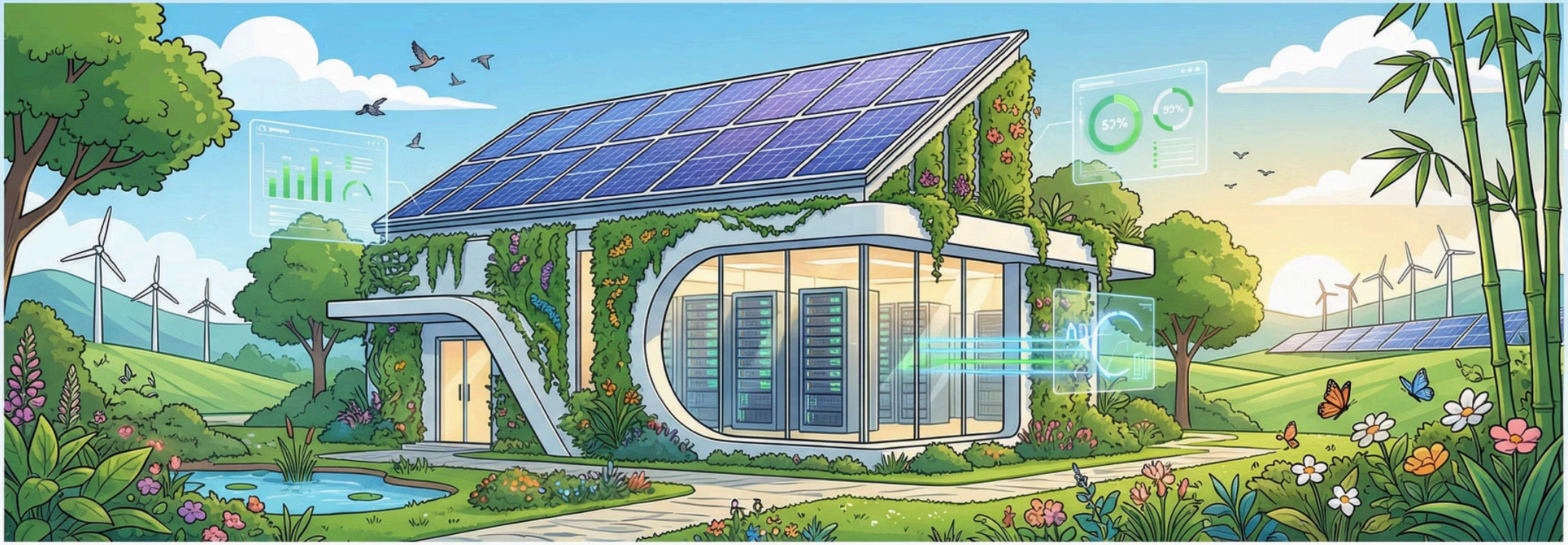
国家政策双碳战略：2030年碳达峰，2060年碳中和全，国家霰碳碳中和目标，对策市破跑碳一艘AI鄙区的国家资夯凿任。



绿色AI不是选择，而是技术发展的必然趋势和道德责任

Green AI is not a choice, but an inevitable trend of technological development and a moral responsibility.

未来：绿色智能的新时代



这一未来愿景展示了技术与自然的完美和谐。我们致力于零碳运营，利用可再生能源驱动计算机能力，为人类服务而不造成环境代价。可持续人工智能的发展是我们实现绿色智能新时代的核心目标。